

Physics-guided placement of conductive particles in a polydisperse packed bed

Ali Khalifa and

Technical University of Munich, Freising, Germany

Abstract

A granular bed containing a limited amount of conductive material can have different thermal performance depending on where that material is placed. We study this placement effect without changing the packing geometry, porosity, particle-size distribution or conductive-material volume. Ten independent 500-particle packings were generated by the discrete-element method at a nominal porosity of 0.38. A steady contact-network model was then used to compare random placement, contact-degree ranking, a heat-path ranking and a fixed-budget swap search. Every heterogeneous case contained exactly 10, 30 and 10 conductive particles from the 1.5, 2.0 and 2.5 mm size classes, respectively. Across the ten packings, random placement gave $k_{\text{eff}} = 0.11046 \pm 0.00413 \text{ W m}^{-1} \text{ K}^{-1}$. Degree ranking improved this value by only 2.4% on average, whereas heat-path ranking improved it by 10.5%. The best assignments found by a fixed computational search budget improved effective conductivity by $56.3 \pm 4.1\%$ relative to the random mean; the improvement was between 50.8% and 60.9% in every packing. Independent thermal calculations reproduced all reported best values exactly, with maximum relative energy-balance error below 3.1×10^{-12} . The results show that conductive particles are most useful when they cooperate to form effective hot-to-cold pathways; selecting locally well-connected particles alone is not sufficient. In a representative packing, optimization converted 35 mostly small conductive clusters into five clusters, including two wall-spanning clusters, and increased the fraction of contact heat carried by high-high contacts from 4.6% to 41.3%. The present conclusions concern stationary, contact-dominated beds and do not yet include a validated pore-fluid conduction model.

Keywords: packed bed; granular heat conduction; discrete element method; contact network; material allocation; inverse design

1 Introduction

Heat transfer through a packed bed depends on particle conductivity, particle size, porosity, contact area and the organization of the contact network. Classical theory and thermal

discrete-element models describe how these factors determine heat flow through a specified granular structure [1, 2, 3, 4]. Most such calculations are forward problems: the packing and material distribution are prescribed, and the resulting temperature field or effective conductivity is calculated.

Many engineering beds, however, contain a limited amount of an expensive or specialized conductive phase. In that situation, the practical question is not only how much conductive material is present, but where it should be placed. Random mixing can put conductive particles in regions that carry little net heat. Selecting particles with many contacts may also be inefficient, because a highly connected particle can lie on a side branch rather than on a route that links the hot and cold boundaries.

Inverse design on graph-structured materials has been considered for architected lattices [5]. A packed bed creates a different problem: the graph is not freely drawn or deformed, but is the mechanically admissible contact network produced by particle packing. The geometry remains fixed and only the particle material labels are changed. This gives a controlled question:

At fixed porosity, particle-size distribution and conductive-material volume, can the effective conductivity of a packed bed be increased substantially by placing conductive particles according to the global heat-transfer network?

We address this question using ten independent DEM packings, a verified steady thermal contact network and four increasingly informed allocation strategies. The study deliberately begins without machine learning. A learned graph policy is useful only if the physical allocation effect is reproducible and if a transparent method does not already solve the problem sufficiently well.

Figure 1 summarizes the physical problem. DEM first fixes the particle positions and contacts. The material budget is then allocated without moving any particle or changing the packing porosity and particle-size distribution. The present paper considers heat conduction through solid contacts only; stagnant-fluid conduction is reserved for a later, separately validated extension.

2 Methods

2.1 Controlled DEM packing ensemble

The packings were generated with LAMMPS using a Hertz–Mindlin granular contact model [9]. Each realization contains 500 spherical particles in a cubic domain with periodic

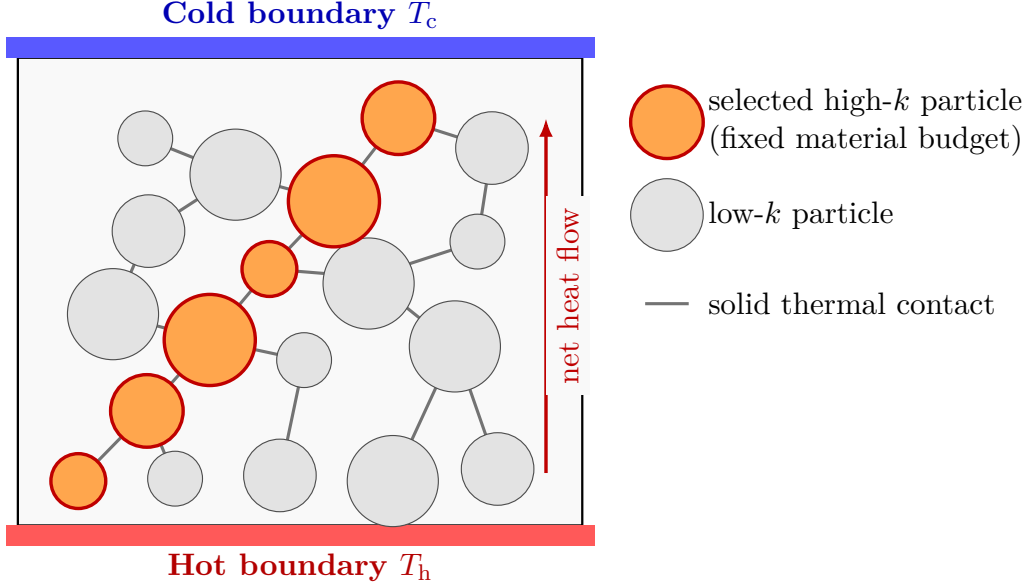


Figure 1: Schematic of the inverse allocation problem for a polydisperse packed bed. The three illustrated particle sizes represent the simulated diameter classes of 1.5, 2.0 and 2.5 mm. DEM fixes the particle positions, particle-size distribution and contact network. A prescribed number of particles from each size class receives the high-conductivity material, while the packing geometry and total conductive-material volume remain unchanged. Heat is conducted through the fixed solid-contact network.

boundaries in the horizontal directions and fixed walls in the vertical direction. The number-based particle-size distribution is identical in every realization: 100 particles of diameter 1.5 mm, 300 of diameter 2.0 mm and 100 of diameter 2.5 mm. The cubic cell size was calculated from the exact undeformed particle volume to impose a nominal porosity of 0.38.

Ten random seeds generated ten different center arrangements and contact networks. All particle counts, sizes, material properties, cell dimensions, growth settings and relaxation settings were held fixed. The resulting contact graphs contained 1325–1382 particle–particle contacts. Every packing had one component that connected the lower and upper walls.

2.2 Steady thermal contact network

Each particle is represented by a graph node and every DEM contact by a thermal edge. For an internal particle i , steady energy conservation is

$$\sum_{j \in \mathcal{N}_i} G_{ij}(T_j - T_i) = 0, \quad (1)$$

where G_{ij} is the contact conductance. For two particles with conductivities k_i and k_j and contact radius a_{ij} , the constriction model is

$$G_{ij} = \frac{2a_{ij}k_ik_j}{k_i + k_j}, \quad a_{ij} = \sqrt{R_{ij}^*\delta_{ij}}, \quad (2)$$

where $R_{ij}^* = R_iR_j/(R_i + R_j)$ and δ_{ij} is the DEM overlap. Fixed temperatures of 301 and 300 K were applied at the lower and upper walls. The effective conductivity follows from

$$k_{\text{eff}} = \frac{QH}{A(T_h - T_c)}, \quad (3)$$

where Q is the wall heat rate, H is bed height and A is cross-sectional area.

The present paper uses solid-contact conduction only. A preliminary stagnant-air gap model was tested, but its predicted conductivity depended too strongly on an arbitrary neighbour cutoff. Those results were therefore excluded from the allocation conclusions. This choice makes the present scope a stationary, contact-dominated bed rather than a flowing or convective system.

2.3 Fixed material budget

The low- and high-conductivity materials were assigned $k_L = 1$ and $k_H = 10 \text{ W m}^{-1} \text{ K}^{-1}$. A binary variable x_i indicates whether particle i receives the high-conductivity material:

$$k_i = k_L + x_i(k_H - k_L). \quad (4)$$

Because the bed is polydisperse, selecting the same total number of particles does not automatically select the same material volume. Every allocation was therefore constrained to contain 10 small, 30 medium and 10 large high-conductivity particles. The selected conductive-particle volume was $2.2514747 \times 10^{-7} \text{ m}^3$ in every calculation. This constraint preserves both the conductive-particle count and its size distribution.

2.4 Allocation strategies

Four strategies were compared on every fixed packing:

1. **Random placement:** 100 independent assignments satisfying the size-class quotas. Their mean and standard deviation define the undesigned reference.
2. **Degree ranking:** within each size class, particles with the most contacts are selected first.

3. **Heat-path ranking:** the homogeneous low-conductivity bed is solved first. Each particle receives the score

$$S_i = \sum_j |G_{ij}(T_i - T_j)|, \quad (5)$$

which measures the baseline heat passing through its incident contacts. The highest-scoring particles are selected within each size class.

4. **Fixed-budget single-swap search:** the search starts from the better of degree and heat-path ranking. One selected and one unselected particle of the same size class are exchanged. The exchange is retained only when it increases k_{eff} . Five independent restarts with 3000 proposed swaps each were used.

The swap search is a simple local-search method based on exchanging two particles. In each proposal, one high-conductivity particle and one low-conductivity particle from the same size class exchange their material labels. This preserves the number, size distribution and total volume of the high-conductivity particles. The new allocation is accepted only if it increases k_{eff} . Similar exchange strategies are widely used in combinatorial optimization [6, 7], although our implementation is adapted specifically to the packed-bed thermal problem. Because only a fixed number of exchanges is tested, the method identifies a good allocation but does not guarantee the global optimum. Performance is evaluated relative to the mean result obtained from random allocations in the same packing:

$$I = 100 \frac{k_{\text{eff}} - \bar{k}_{\text{eff,random}}}{\bar{k}_{\text{eff,random}}} [\%]. \quad (6)$$

2.5 Numerical verification and statistical reporting

The sparse solver was tested for energy conservation and linear scaling. On a representative packing, doubling all particle conductivities doubled both heat rate and k_{eff} . Increasing the imposed temperature difference by a factor of ten increased heat rate by ten while leaving k_{eff} unchanged. Temperatures remained between the boundary values.

The ten DEM realizations are treated as independent packing samples. Results are reported as mean $\pm$ sample standard deviation. For the improvement over random placement, a 95% confidence interval was calculated with the Student t distribution. Each best particle-ID assignment was also passed to an independent call of the thermal solver.

The complete computational workflow is shown in Fig. 2. DEM is required only once for each packing. The same contact graph can subsequently be evaluated for many material

allocations, which makes controlled comparison and fixed-budget search possible without regenerating the packing.

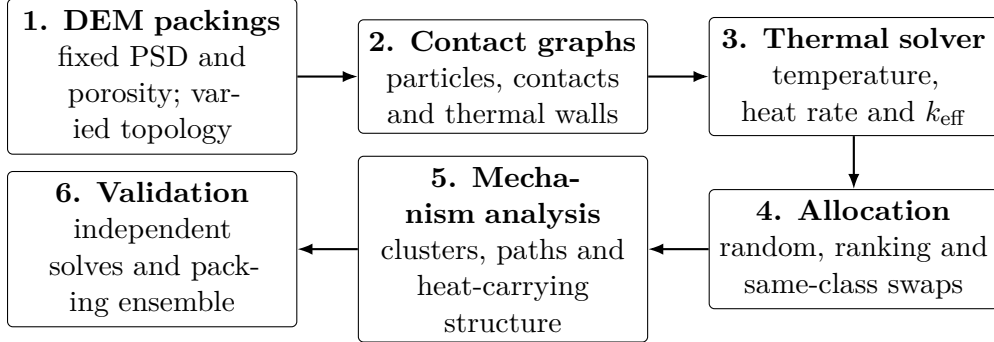


Figure 2: Computational workflow. DEM is run once per packing, whereas the thermal network is repeatedly solved on the same contact graph to compare controlled material allocations and explain their performance.

3 Results

3.1 Homogeneous packing variability

With all particles assigned $k = 1 \text{ W m}^{-1} \text{ K}^{-1}$, the ten packings gave

$$k_{\text{eff,hom}} = 0.09329 \pm 0.00351 \text{ W m}^{-1} \text{ K}^{-1}. \quad (7)$$

The range was $0.08615\text{--}0.09717 \text{ W m}^{-1} \text{ K}^{-1}$ and the coefficient of variation was 3.77%. The mean coordination number was 5.393 ± 0.066 , with a smaller coefficient of variation of 1.22%. Thus, packings with nearly identical contact counts still showed measurable thermal variation, indicating that contact arrangement and conductance weights matter.

Between one and eleven particles per packing were outside the wall-anchored contact component. They carry no heat in the contact-only model and were excluded from the linear system. Every packing nevertheless contained one large component spanning both thermal walls.

3.2 Allocation performance across ten packings

Table 1 shows the ensemble results. Adding 10% conductive particles at random raised the mean conductivity to $0.11046 \pm 0.00413 \text{ W m}^{-1} \text{ K}^{-1}$. Degree ranking produced only a small additional improvement and was slightly worse than the random mean in one packing. Heat-path ranking was better than random in all ten packings.

Table 1: Performance across ten independent DEM packings. Improvement is calculated relative to the controlled random mean in the same packing.

Allocation	k_{eff} [W/(m K)]	Mean improvement
Random mean	0.11046 ± 0.00413	reference
Degree ranking	0.11310 ± 0.00425	$2.4 \pm 2.5\%$
Heat-path ranking	0.12208 ± 0.00467	$10.5 \pm 2.3\%$
Swap search, five-restart mean	0.16868 ± 0.00692	$52.7 \pm 3.5\%$
Best fixed-budget search	0.17263 ± 0.00787	$56.3 \pm 4.1\%$

The best fixed-budget search improved on random placement in every realization. The smallest improvement was 50.8% and the largest was 60.9%. The mean improvement was 56.3%, with a 95% confidence interval of 53.4–59.2%. The corresponding improvement over the homogeneous low-conductivity bed was $85.1 \pm 4.9\%$.

Table 2: Per-packing controlled allocation results.

Seed	Random mean	Heat path	Best search	Gain [%]
18427	0.11123	0.12186	0.17895	60.9
29173	0.10194	0.11045	0.15600	53.0
37811	0.11317	0.12510	0.18102	60.0
46549	0.11216	0.12153	0.16918	50.8
55291	0.11330	0.12602	0.18219	60.8
64109	0.10562	0.12018	0.16895	60.0
72869	0.11075	0.12669	0.17540	58.4
81647	0.11472	0.12467	0.17498	52.5
90439	0.11402	0.12369	0.17334	52.0
99223	0.10774	0.12058	0.16628	54.3

The mean of the five final swap-search values was also substantially above random in every packing, with improvements between 48.7% and 58.1%. This shows that the conclusion does not depend only on selecting the best of five search trajectories.

3.3 Independent reproduction and conservation

The best ID list from every packing was independently evaluated by the thermal solver. All ten values reproduced the allocation-driver results exactly. Every independent calculation contained exactly 50 high-conductivity particles and the original DEM contact graph. The maximum relative difference between hot- and cold-side heat rates was 3.04×10^{-12} . These checks rule out output rounding, a changed graph or a mismatch in particle IDs as explanations for the observed improvement.

3.4 Mechanism comparison in a representative packing

To examine why the optimized assignment performs better, seed 18427 was analysed in greater detail. Three allocations were compared on exactly the same contact graph: a random realization closest to the random mean, heat-path ranking and the best fixed-budget swap-search result. The diagnostic quantities in Table 3 describe connections between high-conductivity particles, their contact with the thermal walls, and the heat carried by the resulting network.

The indicators describe different, complementary aspects of the conductive structure. Their definitions and physical meanings are as follows.

- **Effective conductivity**, k_{eff} , is the overall thermal performance of the complete packed bed. It is calculated from the total heat rate between the hot and cold walls, the imposed temperature difference and the bed dimensions. Unlike the remaining indicators, it includes the combined effect of all contacts and all parallel heat-transfer routes.
- **High-high contacts** counts direct particle contacts for which both particles have the high conductivity. A larger number means that the conductive material is less isolated and has more opportunities to form continuous low-resistance routes. The count alone is not sufficient, however, because contacts on horizontal side branches may contribute little to heat transfer between the two walls.
- **High- k clusters** are connected components formed using only high-conductivity particles and high-high contacts. Two conductive particles belong to the same cluster when they are connected directly or through a chain of other conductive particles. Many clusters therefore indicate a fragmented conductive phase, whereas fewer clusters indicate that the same 50 particles have been consolidated into larger connected structures.
- **Largest high- k cluster** is the number of particles in the largest of these connected components. It shows whether a substantial part of the available conductive material participates in one cooperative structure. A large cluster is useful only if it is located and oriented so that it helps connect the thermal boundaries.
- **Wall-spanning high- k clusters** counts conductive clusters that contain at least one particle touching the hot wall and at least one particle touching the cold wall. Such a cluster provides an uninterrupted sequence of high-conductivity particles across the bed. A value larger than one indicates multiple parallel conductive routes rather than a single chain.

- **High- k particles linked to the hot/cold wall** gives the number of conductive particles in direct thermal contact with each boundary. These particles act as entry and exit points for heat. The notation 7/6, for example, means that seven high- k particles touch the hot wall and six touch the cold wall. High values at only one wall do not by themselves guarantee a complete hot-to-cold route.
- **Heat on high-high contacts** is the sum of the absolute heat rates through all high-high contacts divided by the corresponding sum over all particle-particle contacts. Absolute values are used so that heat rates in different edge directions do not cancel. This fraction measures how much of the active contact-network heat transport actually uses connections between conductive particles. It is therefore more informative than merely counting high-high contacts.
- **Heat-weighted vertical alignment** describes whether the high-high contacts that carry heat are oriented in the direction between the hot and cold walls. For each high-high contact, the alignment is $|\Delta z|/d_{ij}$, which is zero for a horizontal contact and one for a perfectly vertical contact. The reported mean is weighted by the absolute heat rate of each contact. A value closer to one therefore means that the most active conductive contacts are preferentially aligned with the imposed temperature gradient.
- **Minimum-path resistance** is obtained by assigning every thermal link the resistance $1/G_{ij}$ and finding the hot-to-cold path with the smallest sum of link resistances. A smaller value identifies a more favourable single route through the contact graph. It is a diagnostic path measure, not the total bed resistance, because the full thermal solution also includes many parallel and interconnected routes.
- **High- k fraction on the minimum path** is the percentage of particles on that lowest-resistance path that have high conductivity. A value of 100% means that every particle along the identified particle sequence is conductive. This indicator shows whether the material allocation reinforces the most favourable individual route, while k_{eff} remains the measure of the performance of the entire network.

The indicators should consequently be interpreted together. Contact and cluster counts describe how the conductive phase is connected; wall contact and vertical alignment describe whether this connection is useful for the imposed heat-flow direction; and the heat fractions and path resistance show whether the resulting structure is actually used by the thermal solution.

Table 3: Mechanism comparison for packing seed 18427. The minimum-path resistance is a descriptive series-path indicator and not the total resistance of the parallel contact network.

Indicator	Random	Heat path	Optimized
k_{eff} [W/(m K)]	0.1112	0.1219	0.1789
High-high contacts	17	27	55
Number of high- k clusters	35	26	5
Largest high- k cluster [particles]	6	5	26
Wall-spanning high- k clusters	0	0	2
High- k particles linked to hot/cold wall	4/2	0/0	7/6
Heat on high-high contacts [%]	4.6	9.1	41.3
Heat-weighted vertical alignment	0.666	0.728	0.804
Minimum-path resistance [K/W]	19894	15884	2892
High- k fraction on minimum path [%]	63.6	55.6	100

The random allocation divided the 50 conductive particles into 35 mostly small clusters, and no all-conductive cluster connected the two walls. Heat-path ranking increased the number and vertical alignment of high-high contacts, but it also produced no wall-spanning conductive cluster. In contrast, the swap search consolidated the conductive phase into five clusters. The largest contained 26 particles and two clusters connected the hot and cold boundaries.

This structural change also altered where heat travelled. The fraction of the absolute particle-contact heat rate carried by high-high contacts increased from 4.6% for the representative random allocation to 41.3% after optimization. The heat-weighted vertical alignment of those contacts increased from 0.666 to 0.804. Moreover, all nine particles on the calculated minimum-resistance series path were conductive, and its resistance was 85.5% lower than for the random allocation. These observations support a concrete mechanism: the optimizer forms several vertically useful, wall-connected conductive routes and reinforces the pre-existing heat-carrying backbone. The mechanism analysis currently concerns one representative packing and must be repeated for the complete ensemble before it is treated as a general result.

4 Discussion

The main result is simple: the same amount of conductive material can perform very differently depending on its position in the contact network. The weak performance of degree ranking explains why a purely local rule is insufficient. A particle may have many neighbours but still lie in a region with little net vertical heat flow. Heat-path ranking uses the direction and magnitude of the baseline heat flow and therefore performs better. Its consistent 8.3–14.4% improvement is already a useful transparent result.

The much larger gain from the swap search shows that conductive particles must be selected cooperatively. Changing one particle modifies the temperature and heat-flow distribution, which changes the usefulness of other particles. The representative mechanism comparison makes this interpretation more specific: the optimized assignment created two wall-spanning conductive clusters, strongly increased the heat carried by high–high contacts and produced a fully conductive minimum-resistance path. The design problem is therefore global: good assignments build several low-resistance routes between the thermal walls while avoiding isolated conductive clusters and redundant side branches.

The study differs from a conventional conductivity-prediction exercise. The thermal equations remain the evaluator and the goal is to change a discrete material assignment under a strict budget. It also differs from inverse design of freely generated lattices [5]: the contact topology here is mechanically produced by DEM and remains fixed during allocation.

Several limitations define the present scope. First, the contact conductance uses a Hertzian radius derived from DEM overlap, so stiffness and packing pressure can affect the predicted magnitude. Second, the bed contains no validated fluid-phase conduction, convection or radiation. Third, only one conductive fraction and one conductivity contrast have been studied. Fourth, ten packings establish a strong pilot ensemble but remain too few for broad claims about all granular materials. Finally, the swap search is budgeted and stochastic; the reported assignments are the best found, not proven global optima.

5 Current conclusions

Ten equal-porosity, equal-PSD DEM packings were used to isolate the influence of conductive-particle placement. With an identical 10/30/10 size-class budget, degree ranking improved effective conductivity only slightly, while heat-path ranking gave a consistent mean improvement of 10.5% over random placement. The best fixed-budget swap searches improved conductivity by 50.8–60.9% in all ten packings. The effect therefore

survives changes in random packing topology and cannot be attributed to conductive volume or particle-size bias. Exact independent reproduction and near-machine-precision energy conservation support the numerical reliability of the result. In the representative packing, optimization also produced two wall-spanning conductive clusters and increased the high–high contact heat fraction from 4.6% to 41.3%. This provides initial evidence that the improvement results from connected, vertically useful conductive pathways rather than from high local coordination alone.

6 Next steps

The work should now proceed in the following order so that each step answers one clear question before additional complexity is introduced.

1. **Complete the mechanism analysis across the ensemble.** Repeat the cluster, wall-connection, vertical-alignment, heat-fraction and minimum-path calculations for all ten packings. Determine which indicators consistently distinguish optimized from random and heat-path allocations, and report their packing-to-packing variability.
2. **Test robustness of the main effect.** Repeat the ten-packing comparison for at least one lower and one higher conductive fraction, for example 5% and 20%, while preserving size-class volume fractions. Test at least one additional conductivity ratio. The aim is to identify where placement matters strongly and where it becomes unimportant.
3. **Check contact-model sensitivity.** Repeat selected cases with a changed particle stiffness or an alternative normalized contact/interfacial- resistance law. The ranking of allocation strategies should remain stable even if the absolute conductivity changes.
4. **Add the fluid phase only with a defensible model.** Replace the provisional distance-cutoff gap network with a geometry-defined pore network, preferably based on a radical Voronoi or weighted-Delaunay construction for polydisperse spheres. Recheck whether the placement benefit remains when stagnant-fluid conduction provides parallel paths.
5. **Decide whether graph learning is necessary.** A graph neural network is justified only if it can reproduce high-quality allocations on unseen packings much faster than repeated thermal solves, or if it transfers useful placement rules across particle counts and operating conditions. Graph networks are appropriate for such relational data [8], but the transparent heat-path and swap baselines must remain mandatory.

6. **Expand only after the design is fixed.** Increase the packing ensemble for final statistical analysis and, if graph learning is pursued, construct training, validation and test splits at the packing level so that allocations from the same contact graph never appear in different subsets.

References

- [1] G. K. Batchelor and R. W. O’Brien, “Thermal or electrical conduction through a granular material,” *Proceedings of the Royal Society of London A*, vol. 355, pp. 313–333, 1977.
- [2] W. L. Vargas and J. J. McCarthy, “Heat conduction in granular materials,” *AIChE Journal*, vol. 47, no. 5, pp. 1052–1059, 2001. doi:10.1002/aic.690470511.
- [3] M. Kiani-Oshtorjani and P. Jalali, “Thermal discrete element method for transient heat conduction in granular packing under compressive forces,” *International Journal of Heat and Mass Transfer*, vol. 145, 118753, 2019.
- [4] C. Joulin, A. Xiang, P. D. Lee and C. E. Krill III, “Capturing heat transfer for complex-shaped multibody contact problems using the discrete element method,” *Computational Particle Mechanics*, 2020. doi:10.1007/s40571-020-00321-w.
- [5] D. Dold and D. Aranguren van Egmond, “Differentiable graph-structured models for inverse design of lattice materials,” *Cell Reports Physical Science*, vol. 4, no. 10, 101586, 2023. doi:10.1016/j.xcrp.2023.101586.
- [6] B. W. Kernighan and S. Lin, “An efficient heuristic procedure for partitioning graphs,” *The Bell System Technical Journal*, vol. 49, no. 2, pp. 291–307, 1970. doi:10.1002/j.1538-7305.1970.tb01770.x.
- [7] D. Maringer and H. Kellerer, “Optimization of cardinality constrained portfolios with a hybrid local search algorithm,” *OR Spectrum*, vol. 25, pp. 481–495, 2003. doi:10.1007/s00291-003-0139-1.
- [8] P. W. Battaglia et al., “Relational inductive biases, deep learning, and graph networks,” *arXiv preprint arXiv:1806.01261*, 2018. doi:10.48550/arXiv.1806.01261.
- [9] LAMMPS Developers, “`pair_style granular` command,” LAMMPS documentation. docs.lammps.org/pair_granular.html.